MIL-C-55486B(EL)
15 May 1969
Superseding
MIL-C-55486A(EL)
2 May 1967

MILITARY SPECIFICATION

CABLE ASSEMBLY TELEPHONE CX-4566( )/G and CX-4760( )/U

1. SCOPE

1.1 Scope.- This specification covers Cable Assemblies CX-4566( )/G and CX-4760( )/U. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-M-13231	Marking of Electronic Items.
MIL-C-55074	Connector, Plug, Electrical U-185( )/G.
MIL-C-55442	Packaging and Packing of Cable Assemblies and Cord Assemblies.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment.

FSC 5995

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DRAWINGS

ELECTRONICS COMMAND

SC-DL-34030	Cable Assembly CX-4566( )/G.
SC-DL-40468	Cable Assembly CX-4760( )/U.
SC-GL-323007	Electrical Fixture for Connectors U-185( )/G, U-186( )/G, U-187( )/G and Contact Assembly MX-3227( )/G.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Cable assemblies covered by this specification are as follows:

3.1.1 Cable Assembly CX-4566( )/G (_FT_IN).- Consists of a length of Cable WM-130( )/G with a Connector, Plug, Electrical U-185( )/G on each end.

3.1.2 Cable Assembly CX-4760( )/U.- Consists of a 15 foot length of Cable WM-130( )/G with a Connector, Plug, Electrical U-185( )/G on one end, the other end is stripped and fanned. (See 6.4).

3.2 Construction.- Cable assemblies shall be constructed in accordance with the requirements of this specification and the following Drawing and Data Lists:

(a) Drawing and Data List SC-DL-34030 for Cable Assembly CX-4566( )/G.

(b) Drawing and Data List SC-DL-40468 for Cable Assembly CX-4760( )/U.

3.3 Cleaning.- After assembling, the equipment shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

3.4 Marking.- Marking shall conform to MIL-M-13231. (See 4.3).

3.5 Connectors.- Connector, Plug, Electrical U-185( )/G shall meet the requirements of MIL-C-55074, except that the Pull, Moisture resistance, and High humidity and high temperature tests shall be performed as part of this specification. (See 4.11).

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3.6 Electrical requirements.-

3.6.1 Continuity.- The resistance values shall be in accordance with the following: (See 4.7.1).

Equipment	
CX-4566( )/G (250 ft)	10.0 ohms max.
CX-4566( )/G (25 ft)	1.0 ohm max.
CX-4760( )/U	1.0 ohm max.

3.6.2 Dielectric strength.- The cable assembly shall be considered a reject if the automatic tester fails to proceed after a period of 1 minute when tested in accordance with 4.7.2.

3.6.3 Insulation resistance.- When tested in accordance with 4.7.3, the insulation resistance shall be not less than 1,000 megohms except during the Moisture resistance test 4.9.1 and the High humidity and high temperature test 4.9.2 when it shall be not less than 10 megohms. (See 4.6).

3.7 Mechanical requirements.-

3.7.1 Immersion.- After being subjected to the tests specified in 4.8.1, each cable assembly shall meet the requirements of 3.6.2 Dielectric strength and 3.6.3 Insulation resistance, and there shall be no evidence of water in the connector housings.

3.7.2 Pull.- After being subjected to the test specified in 4.8.2, the cable assemblies shall show no straining of individual conductors or penetration of the grip to the steel braid of the cable and shall meet the requirements of 3.6.1 Continuity, 3.6.2 Dielectric strength, 3.6.3 Insulation resistance and 3.7.1 Immersion.

3.8 Service conditions.-

3.8.1 Moisture resistance.- When subjected to the test specified in 4.9.1, the cable assemblies shall meet the requirements of 3.6.3 Insulation resistance, and there shall be no evidence of corrosion.

3.8.2 High humidity and high temperature.- When subjected to the test specified in 4.9.2, the cable assemblies shall meet the requirements of 3.6.3 Insulation resistance, and there shall be no evidence of corrosion.

3.9 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraph: (See 4.10).

3.3 Cleaning

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents. (See 4.3).

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.4).

(2) Quality conformance inspection of preparation for delivery. (See 4.12).

4.3 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where required
Marking	3.4

4.4 Quality conformance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

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4.4.1 Group A inspection.- Each unit (100%) of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQLs indicated in table I. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table I - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Electrical				
Continuity (Cable Assembly)	3.6.1	4.7.1)	.65% inspection	
Dielectric strength (Cable Assembly)	3.6.2	4.7.2)	by automatic	
Insulation resistance (Cable Assembly)	3.6.3	4.7.3)	means	
Visual and mechanical	3.9	4.10	1.0%	4.0%

4.4.2 Group B inspection.- This inspection, including sampling, shall conform to table II and to the special procedures for small-sample inspection of MIL-STD-105. The inspection level shall be S-3 for normal, tightened and reduced inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be as shown below.

Table II - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Immersion (Cable Assembly)	3.7.1	4.8.1	1.5% defective

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4.4.3 Group C inspection.- This inspection shall consist of the tests specified in table III, table IV and table V and shall be performed on samples that have been subjected to and met group A and group B inspection. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of production shall be selected from the first units produced. Regardless of nomenclature or cable length, each item on the contract shall be considered a unit for purposes of the group C tests.

4.4.3.1 Group C-1 inspection.- This inspection shall consist of the test specified in table III and shall be performed on 2 units every month or every 1,000 units, whichever comes first.

Table III - Group C-1 inspection

Inspection	Req. Para.	Insp. Para.
Pull (Cable Assembly)	3.7.2	4.8.2

4.4.3.2 Group C-2 inspection.- This inspection shall consist of the test specified in table IV and shall be performed on 4 units every 3 months or every 3,000 units, whichever comes first.

Table IV - Group C-2 inspection

Inspection	Req. Para.	Insp. Para.
High humidity and high temperature (Cable Assembly)	3.8.2	4.9.2

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4.4.3.3 Group C-3 inspection.- This inspection shall consist of the test specified in table V and shall be performed on 4 units once each year or every 12,000 units, whichever comes first.

Table V - Group C-3 inspection

Inspection	Req. Para.	Insp. Para.
Moisture resistance (Cable Assembly)	3.8.1	4.9.1

4.4.3.4 Noncompliance.- The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, the final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5 Automatic testing.- The inspection for Continuity 4.7.1, Dielectric strength 4.7.2 and Insulation resistance 4.7.3 shall be performed on all equipments by automatic means using electrical fixtures specified on SC-GL-323007. Automatic test equipment other than that referenced on SC-GL-323007 shall have prior Government approval. The automatic test equipment as specified is programed to test each terminal in turn with the remaining terminals connected together to ground for dielectric strength and insulation resistance simultaneously. The automatic test equipment is also programed to test for continuity sequentially--terminals 1A thru 26A, IB thru 26B with the read out on the tester showing 1 thru 52.

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4.6 Preconditioning.- The contractor shall be permitted to precondition the cable assemblies prior to performing any electrical tests except the electrical test performed during the Moisture resistance test 4.9.1 and the High humidity and high temperature test 4.9.2. The preconditioning shall consist of removing surface moisture from the equipment by wiping, drying, blowing or heating. The application of heat shall be limited to 3 minutes. For those equipments subjected to Immersion 4.8.1, Moisture resistance 4.9.1 and High humidity and high temperature 4.9.2 unmated and without protective covers, a 24 hour waiting period in an ambient room temperature is permitted. However, if the contractor exercises this option of preconditioning, the contractor shall precondition all samples selected by Quality Assurance Representatives immediately prior to testing.

4.7 Electrical tests.-

4.7.1 Continuity.- Electrical continuity of the cable assemblies shall be verified by a point-to-point test by automatic means (See 4.5) to determine conformance with the requirements of 3.6.1. The measurements include the jumper wires of the contact assembly.

4.7.2 Dielectric strength.- The cable assemblies shall be tested by automatic means. (See 4.5). The 1,000 volts supplied by the tester shall be applied instantaneously and the sequence of test shall be as fast as the tester will permit. The cable assembly shall meet the requirements of 3.6.2.

4.7.3 Insulation resistance.- The cable assemblies shall be tested by automatic means. (See 4.5). The test will be performed with a voltage of 1,000 volts as supplied by the tester, except during the Moisture resistance test 4.9.1 and High humidity and high temperature test 4.9.2 the voltage shall be 100 volts. The cable assembly shall meet the requirements of 3.6.3.

4.8 Mechanical tests.-

4.8.1 Immersion.- The connectors on the cable assemblies shall be immersed to a depth of 3 feet of water for 16 hours to determine compliance with the requirements of 3.7.1. Fifty percent shall be mated, twenty-five percent shall be with protective cover in place and twenty-five percent shall be without protective cover.

4.8.2 Pull.- One connector on each of 2 cable assemblies shall be subjected to a 800 pound static load for 16 hours followed by twenty-five 200 pound pulls and relaxations. Mating of connectors is permitted. A minimum of 3 feet of cable of each assembly shall be subjected to the load. The cable assemblies shall meet the requirements of 3.7.2.

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4.9 Service tests.-

4.9.1 Moisture resistance.- Four cable assemblies, 2 mated, 1 with protective cover in place and 1 without protective cover shall be subjected to 15 continuous 48-hour cycles per MIL-STD-170. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to the standard. Insulation resistance measurements shall be made during the periods shown on the standard. The equipment shall meet the requirements of 3.8.1.

4.9.2 High humidity and high temperature.- Four cable assemblies, 2 mated, 1 with protective cover in place and 1 without protective cover shall be placed in a chamber at 86°F ± 5° and a relative humidity of 92 to 98 percent. The temperature and humidity shall be maintained continuously for 5 days. While still at these conditions, the insulation resistance shall be measured on the last day. The equipment shall meet the requirements of 3.8.2.

4.10 Visual and mechanical inspection.- The cable assemblies shall be examined for defects listed in table VI. (See 3.9).

4.11 Connectors.- Connector, Plug, Electrical U-185( )/G shall be tested prior to fabrication of cable assemblies to determine conformance with the requirements of 3.5.

4.12 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-C-55442(EL) to determine conformance to the requirements of section 5.

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Table VI - Classification of visual and mechanical defects

Classification	Defect
Major	Locking lug chipped or broken. Locking ring chipped or broken. Threads stripped. Rubber boot cut or ripped. Sharp edges on housing. Epoxy resin on contacts or terminals. Locking ring not operating freely. Housing chipped or broken. Contacts missing, broken, or bent. Contact assembly chipped or broken. Any foreign objects or material preventing the mating of connectors. Cuts or abrasions on gaskets. Cuts on cable. Missing identification band. Wire grip missing. Scratches, cuts, abrasions, etc. with exposure of bare metal.
Minor	Screws missing. Finish not as specified. Marking incorrect. Scratches, cuts, abrasions, etc. without exposure of bare metal. Finish adhesion-flaking, peeling or chipping. Scratches, cuts, abrasions, etc. of contact assembly. Caps not properly secured to block. Boot not properly bonded to gland. Abrasions on cable.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be as specified in MIL-C-55442(EL). (See 6.2(c)).

6. NOTES

6.1 Intended use.- The cable assemblies covered by this specification are intended for use in military communication systems. (See 1.1).

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Levels of preservation, packaging, packing and marking. (See section 5).

(d) Marking and shipping of samples.

(e) Place of final inspection.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example, CX-4566W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Cable Assembly CX-4760( )/U.- In order to expedite testing of Cable Assembly CX-4760( )/U, the contractor may manufacture the assembly in approximately 30 foot lengths similar to Cable Assembly CX-4566( )/G. These assemblies shall be tested as CX-4566( )/G and following submissions and acceptance by the Government QARIC the assemblies shall be returned to the contractor. These assemblies shall then be cut into 2 lengths, stripped and fanned. Then the completed CX-4760( )/U shall be returned to the Government QARIC for inspection of the "fanned" end. At his discretion the Government QARIC may reinspect for electrical requirements.

CUSTODIAN

ARMY EL

Preparing Activity

ARMY EL

Project No. 5995-A078

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